Flight Delay Prediction via Gradient Boosting and Learned Categorical Embeddings

Team T001

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Abstract

This project predicts arrival delays of at least 15 minutes using information available two hours before scheduled departure. Logistic Regression and XGBoost are evaluated on chronological partitions of 2013 New York flight records, with an embedding multilayer perceptron (MLP) as a supplementary comparison. The test set contains 48,581 flights. XGBoost achieves average precision (AP) of 0.3722, ROC-AUC of 0.6367 and Brier loss of 0.1910; the MLP achieves the highest AP, 0.3936. The implementation includes time-safe historical features, development-based threshold selection and an offline schedule-shift demonstration.

I Problem and Related Work

The objective is to estimate whether a flight will arrive substantially late before it departs, then compare small schedule changes using the fitted model. The prediction timestamp is fixed at scheduled departure minus two hours. Current-flight actual departure, arrival, departure delay and airborne time are excluded.

Rebollo and Balakrishnan study the characterization and prediction of air-traffic delays [1]. Lambelho et al. connect delay and cancellation prediction to strategic schedule assessment [2]. These studies motivate combining operational attributes with historical patterns; their scores are not directly comparable across datasets and prediction horizons. XGBoost provides a regularized tree-boosting approach [3], while entity embeddings learn compact representations of categorical identities [4]. Calibration research motivates checking probability quality separately from classification ranking [5].

The project applies these ideas under an explicit information boundary: features must be available at the chosen decision time, and later flights must not affect earlier predictions.

II Dataset Quality and Exploration

The dataset contains 336,776 flight schedules from EWR, JFK and LGA in 2013. The underlying flight-record schema is documented by `nycflights13` [6]. Arrival delay is observed for 327,346 rows; 9,430 rows without an outcome are excluded from supervised evaluation rather than assigned an on-time label. The labeled population contains 80,100 delayed and 247,246 non-delayed flights, approximately 24.47% delayed.

Validation checks cover required columns, clock parsing, timestamp consistency and outcome availability. Schedule rows also support scheduled-demand features, including rows without supervised labels. Exploration examines missingness, class balance, scheduled departure hour, month, carrier and airport. Figure 1 summarizes variation with departure hour and motivates time-of-day features. Observed associations guide feature design without establishing causes.

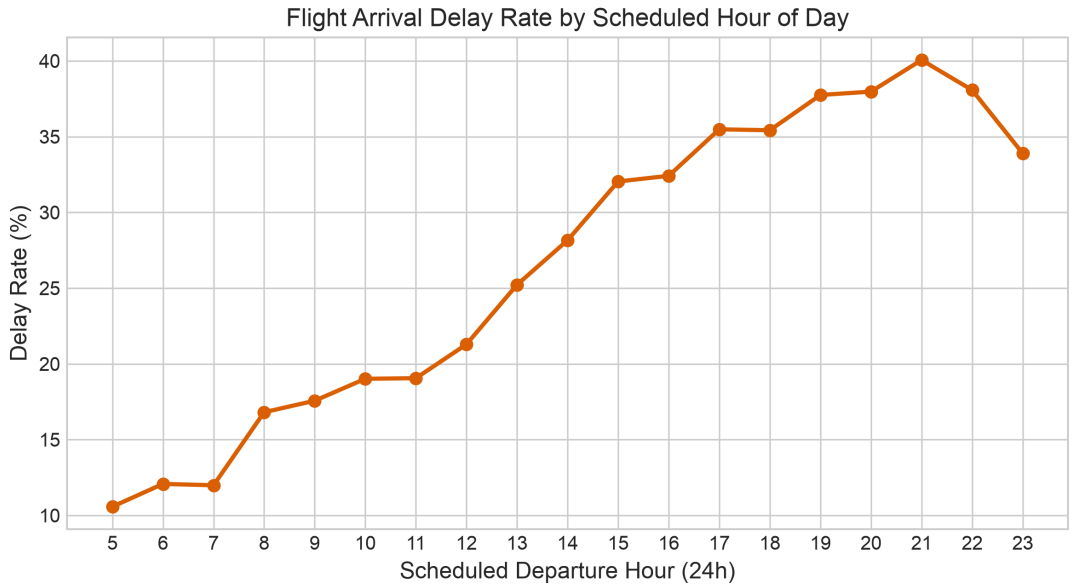


Figure 1: Observed delay frequency by scheduled departure hour.

III Features and Prediction-Time Safety

The target is $y_i = \mathbb{1}[\text{arrival delay}_i \geq 15]$. Features include carrier, origin, destination, distance, calendar variables, scheduled clocks and scheduled origin demand. Cyclic time is encoded as

$$(\sin(2\pi t/P),\ \cos(2\pi t/P))\,, \tag{1}$$

where P is the relevant period. This preserves adjacency across midnight.

Candidate historical features include smoothed carrier, airport and route delay rates and seven-day released-outcome summaries. A past outcome becomes eligible only after actual arrival plus an assumed 24-hour reporting lag, before the local-midnight feature snapshot. This explicit lag models information availability. Daily snapshots are selected using the decision date; automated checks verify that future outcomes cannot alter earlier features.

Preprocessing is fitted on training data. Logistic Regression uses one-hot identities and 28 selected numeric inputs; XGBoost uses the identities and 18 numeric inputs. The MLP uses learned carrier, origin and destination embeddings with the XGBoost numeric set. Feature subsets are selected per model under the same label, date partitions and availability boundary.

IV Models and Experimental Design

A Models and objectives

Logistic Regression supplies an interpretable linear reference. It minimizes binary cross-entropy with L2 regularization:

$$\mathcal{L} = -\frac{1}{N}\sum_i [y_i \log p_i + (1-y_i)\log(1-p_i)] + \lambda \|w\|_2^2. \tag{2}$$

Its selected configuration uses $C = 0.03$ and no class weighting. XGBoost captures nonlinear interactions using 42 trees, maximum depth 3 and learning rate 0.06. The supplementary MLP uses carrier/origin/destination embedding dimensions of 16/16/32, hidden layers of 128, 64 and 32 units, and Adam optimization for 15 fixed epochs. It is a tabular embedding network; no sequence-model result is claimed.

B Chronological validation

Table I gives the 2013 date partitions. Four expanding training folds support temporal selection, with 128 candidate fits. Training outcomes not yet released at the development boundary are purged, leaving 227,520 usable fitting labels from the nominal 228,299 training rows. Development data select probability treatment and the operating threshold. Estimator weights remain fixed during test evaluation; eligible historical inputs can update as outcomes are released. Reported results are retrospective reevaluations of the saved experiment.

Table I: Chronological partitions, all in 2013.

Partition	Dates	Flights
Training	Jan. 1–Sep. 12	228,299
Development	Sep. 13–Nov. 6	50,466
Test	Nov. 7–Dec. 31	48,581

C Evaluation and decision rule

AP summarizes precision–recall performance; ROC-AUC measures ranking across score thresholds. Precision–recall analysis is useful for imbalanced classes [7]. Probability quality is measured by Brier loss,

$$\text{Brier} = \frac{1}{N}\sum_i (p_i - y_i)^2, \tag{3}$$

where lower values are better. Accuracy, precision and recall are computed at the saved operating points, rather than an arbitrary common threshold.

Identity, sigmoid and isotonic probability treatments are compared using development Brier loss. Identity is selected for all three learned models. A project-selected target of 70% precision with at least 100 alerts is not met on the development selection block; the saved policy therefore uses its supported maximum-F1 fallback. The XGBoost threshold is 0.206462. The 70% value is an experimental target, not a faculty-specified requirement or an achieved score.

V Results and Discussion

Table II reports the saved test metrics. The constant-prior reference has AP 0.2686 and ROC-AUC 0.5000. Its 73.14% accuracy accompanies zero delayed-flight recall, illustrating why accuracy alone is insufficient for this task.

XGBoost improves AP and ROC-AUC over Logistic Regression and gives the lowest Brier loss among learned models. The MLP has the highest observed AP and recall. These results describe different ranking, probability and operating-point strengths; they do not identify a single winner on every metric. Figure 2 shows the precision–recall comparison.

Table II: Test results on 48,581 flights. Accuracy, precision and recall use saved development-selected operating points. A dash denotes no positive alerts.

Model	AP ↑	ROC-AUC ↑	Brier ↓	Accuracy	Precision	Recall
Logistic Regression	0.3200	0.5797	0.2396	68.17%	33.99%	19.67%
XGBoost	0.3722	0.6367	0.1910	56.79%	34.49%	67.74%
Embedding MLP	0.3936	0.6506	0.2017	47.94%	31.95%	83.04%
Constant prior	0.2686	0.5000	0.1965	73.14%	—	0.00%

Figure 3 compares predicted probabilities with observed delayed fractions. Raw and selected curves overlap because identity wins probability-treatment selection. This selection does not itself establish perfect calibration. The Brier score and reliability plot therefore remain part of the reported evaluation.

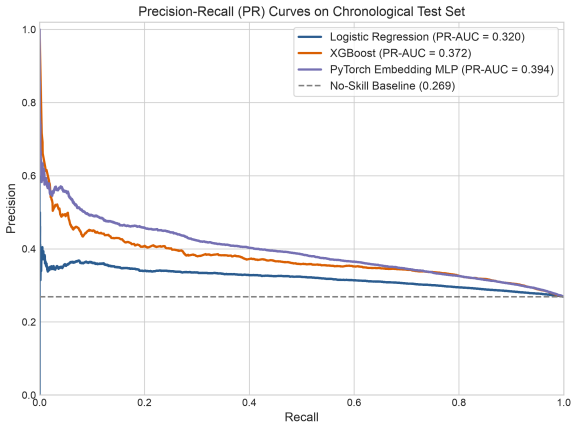


Figure 2: Precision–recall curves from the saved test predictions.

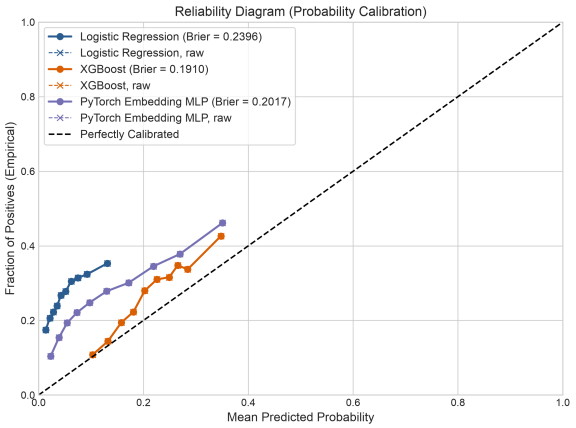


Figure 3: Test reliability assessment for raw and selected probabilities.

VI Application and Offline Rescheduling

The application loads the saved estimator, preprocessing, probability treatment and decision policy to score historical flight inputs. The offline analysis tests departure shifts of −60, −30, 0, +30 and +60 minutes, preserving duration and original decision-time history. Scheduled demand is recomputed against a fixed background timetable. Equal predicted risks favor the smaller absolute shift.

For AA JFK–MIA on 7 November 2013, the original 07:10 departure has predicted risk 10.281%. Both earlier shifts give 10.014%, so the rule selects −30 minutes, or 06:40. The change is −0.267 percentage points, with duration fixed at 195 minutes. This is an offline model-score comparison under explicit assumptions, not evidence that an actual schedule change causes fewer delays.

VII Implementation and Reproducibility

The Python code separates ingestion, time handling, feature construction, temporal experiments, fitting, evaluation, inference and rescheduling. The stack uses pandas, scikit-learn, XGBoost, PyTorch and matplotlib. Saved artifacts include split manifests, selected configurations, thresholds, predictions, metrics and model bundles. The executed notebook provides EDA and analysis; the application demonstrates prediction and schedule comparisons.

Verification includes 37 passing tests, with 74% measured coverage. Tests cover prediction-time leakage, historical release, timestamp handling and inference behavior. Repository documentation records installation, execution commands and modeling assumptions. Results can be traced to `results/metrics/evaluation_metrics.json`, `results/reports/` and the saved model artifacts.

VIII Conclusion

The project implements a reproducible pre-departure prediction workflow with chronological validation and actual model comparisons. XGBoost provides the strongest probability-loss result among learned models; the supplementary embedding MLP provides the highest observed AP. The application extends the evaluated model to an explicit offline schedule-shift analysis.

References

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